

Multimodal Deep Learning

Answer Sheet 7

Date: July 1, 2025

Exercise 1: Selective Fine-Tuning

(a) Fishmask

(i) Mapping

Problem Statement:

- A layer contains 4.0×10^6 weights.
- A "Fisher score approximation" assigns a scalar importance I_i to each weight w_i .
- We are allowed to update only 0.8% of these weights.
- This refers to the Fisher Information Matrix (FIM), which measures how sensitive the loss function is to parameter changes.
 - High Fisher score I_i : important weight.
 - Low Fisher score I_i : less important.
 - One-shot Fisher approximation computes importance without retraining.

Answer:

- Total weights: 4 million parameters.
- Budget: 0.8% of total weights = 32,000 weights = k .
- A binary mask $\mathbf{m} \in \{0, 1\}^n$ has 32,000 ones and the rest zeros. ($m_i = 1$: update; $m_i = 0$: freeze)
- Mapping formula (using both $\mathbf{I}$ and $\mathbf{m}$): $\mathbf{M} = \mathbf{I}^T \mathbf{m}$
- Mapping formula (constructing $\mathbf{m}$ from top- k): $\mathbf{m} = \mathbf{I} > \min(\text{Top}_k(\mathbf{I}))$